

**SIMATS ENGINEERING
ASSESSMENT TOOL**

COURSE NAME: Theory of Computation

COURSE CODE: CSA13

COURSE OUTCOME COVERED

CO2: Design Finite Automata DFA, NFA, ϵ -NFA and demonstrate their equivalence for recognizing regular languages. (BL:3)

Assessment Tool Weightage: 40%

QUESTIONS: 5
Duration : 1 Hour

Total Marks:25

Design Task

Code of Conduct:

I V. Smayika (192372115) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Smayika
Signature of the student:

Regular Expression from the given Automaton

From the automaton:

- $q_0 \rightarrow q_1$ on 0, 1
- $q_1 \rightarrow q_2$ on 0, 1
- $q_2 \rightarrow q_1$ on 0, 1
- $q_2 \rightarrow$ is the final state

The automaton accepts strings of even length greater than or equal to 2.

Regular Expression:

$$(0+1)(0+1)((0+1)(0+1))^*$$

2. Given Grammar

$$S \rightarrow aSc \mid T \mid U \mid \epsilon$$

$$T \rightarrow bTc \mid \epsilon$$

$$U \rightarrow aUbc$$

A) Language Defined

The grammar generates:

$$L = \{a^n b^n\} \cup \{b^n c^n\} \cup \{a^n c^n\}, n \geq 0$$

B) Derivation Tree for aaabbc.Derivation:

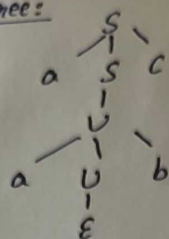
$$S \Rightarrow aSc$$

$$\Rightarrow aaUbc$$

$$\Rightarrow aaabbbc$$

$$\Rightarrow aaabbc$$

Simple tree:



Thus,

$$S \Rightarrow aabbcc$$

c) Is the grammar Ambiguous?

No, the grammar is not ambiguous.

For a given string, the required production (T, U, or recursive S) is determined by its terminal pattern, so there are no two distinct parse trees for the same word.

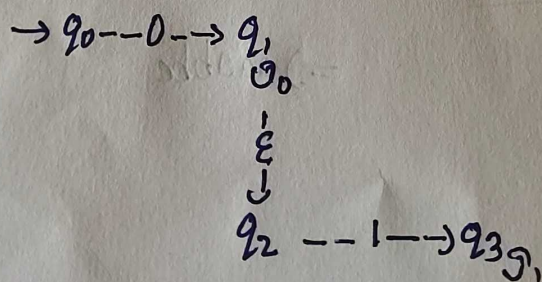
Grammar is unambiguous

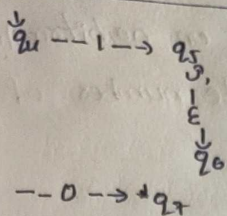
E-NFA for 00^*1+11^*0

Given:

$$00^*1+11^*0$$

An E-NFA can be constructed by connecting the parts E-transitions,





Here:

- 0^+ is represented by the loop on q_4 .
- 1^+ requires at least one 1.
- 1^+ is represented by the loop on q_5 .
- q_7 is the final state.

Closure property and Prime Language

i) If L_1 and L_2 are regular, prove $(L_1 + L_2)$ is regular.

If L_1 and L_2 are regular, there exist finite automata M_1 on M_2 accepting them.

For union:

$$L_1 \cup L_2$$

$\therefore L_1 \cup L_2$ is regular.

ii) Is $\{a^p \mid p \text{ is prime}\}$ regular?

No, it is not regular.

The language contains strings whose lengths are prime

$$a^2, a^3, a^5, a^7, a^{11}, \dots$$

A finite automaton cannot determine whether an arbitrary number is prime because it has only a finite number of states.

Therefore:

$\{0^n \mid n \text{ is prime}\}$ is not regular.

5. DFA Minimization

From the given DFA:

Final states:

q_1, q_2, q_4

Initial state

q_0

The equivalent states are:

$q_0 \equiv q_3$

and

$q_1 \equiv q_2 \equiv q_4$

Thus, the original 6 states can be reduced to 3 states.

Minimized states

Let:

- $A = \{q_0, q_3\} \rightarrow \text{Initial}$
- $B = \{q_1, q_2, q_4\} \rightarrow \text{Final}$
- $C = \{q_5\} \rightarrow \text{Dead / non-final}$

Minimized Transition Table

State	0	1
$\rightarrow A$	A	B
* B	B	C
C	C	C

Result:

Original DFA : 6 states

Minimized DFA : 3 states

RUBRICS FOR CALCULATION

SIMATS ENGINEERING ASSESSMENT TOOL

Criteria	Weightage	Level 4 - Exemplary (5 Marks)	Level 3 - Proficient (4 Marks)	Level 2 - Developing (2-3 Marks)	Level 1 - Beginning (0-1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

94/100